# UI Current Issues

By Christopher Paquin

The UI for our project has a solid foundation but has many indications of multiple developers having various changes without really maintaining a cohesive vision. There are a wide range of issues that I believe should be addressed prior to a high-quality demo for our sponsors. Below are a list of issues and my proposed fixes.

All these fixes can be inspected on THIS branch in Github.

**Global Colors**

Themes use a small palette of colors which keeps things simple but has led to instances of low contrast text being unreadable. Examples are the GPS location on the incident detail screen and the filter button on the ride session screen.

I have added an extra 2 properties called surfaceText and surfaceTextSecondary which allow for more text colors in each palate. I updated all screens/text/themes to use this and the results are that I can no longer find any unreadable text on any screen/theme.

**New Default Theme “Arizona”**

Considering the focus of the Rob Dollar foundation and our school’s location I replaced the old commented out wildcard theme with an Arizona theme which you can see below. I think the other themes have good value as high contrast options but mostly for the demo to appear more “in-theme” I think this does the job for now.

**Screen Size Disparity**

As we saw in the Android Emulator Report, there are big changes in appearance based on what size screen the app is running on. There are large blank spaces above or below boxes. There are also large areas like between the SETTINGS text and the top setting that look off.

I encountered a similar issue on a personal app project and found a similar solution for typescript. I changed many instances of flex to flexgrow which fills predetermined spaces

based on screen size. As you can see from the images below this looks good in every phone screen size. I also tweaked some of the margins and gaps to eliminate open space.

**Image Fade Effect on Ride Session Screen**

The image fade effect was very strong and set to 100% opacity over 40 pixels. On some images that looked far too strong to me, and I toned it down to 90% opacity over 16 pixels. You can see the effect below and I think this is a more balanced setting.

**Other Observations**

In the course of my testing which was far more lengthy than my previous testing, I discovered a number of bugs and situations that I feel warrant discussion.

**Simulated IBDC**

While processing image chunks there is a period of several seconds where no input is registered because the app is purely focused on image chunk processing. This happened constantly through countless application runs. I believe we need should look into having this processing work on a separate thread as the impact to the user is substantial and negative.

The simulator follows production logic of adding new incidents to the most recent session as the time difference is less than 2 hours. Since this works as intended the option to add mock incidents I believe should be expanded to add a deterministic number of sessions and incidents to better populate many sessions rapidly for testing and demonstration. Currently it's a tedious process. It also needs to be able to create old sessions to demonstrate deleting sessions by age.

While the team discussed disconnect logic and in production, I think our current system would be effective, for testing we need a way to disconnect the simulated IBDC as the incident count gets ridiculous after a few minutes. Incidents showing up quickly is great, but we'd have 100+ incidents by the end of the demo. Additionally, if we could vary the image being sent, I think that would be best.

**Interface**

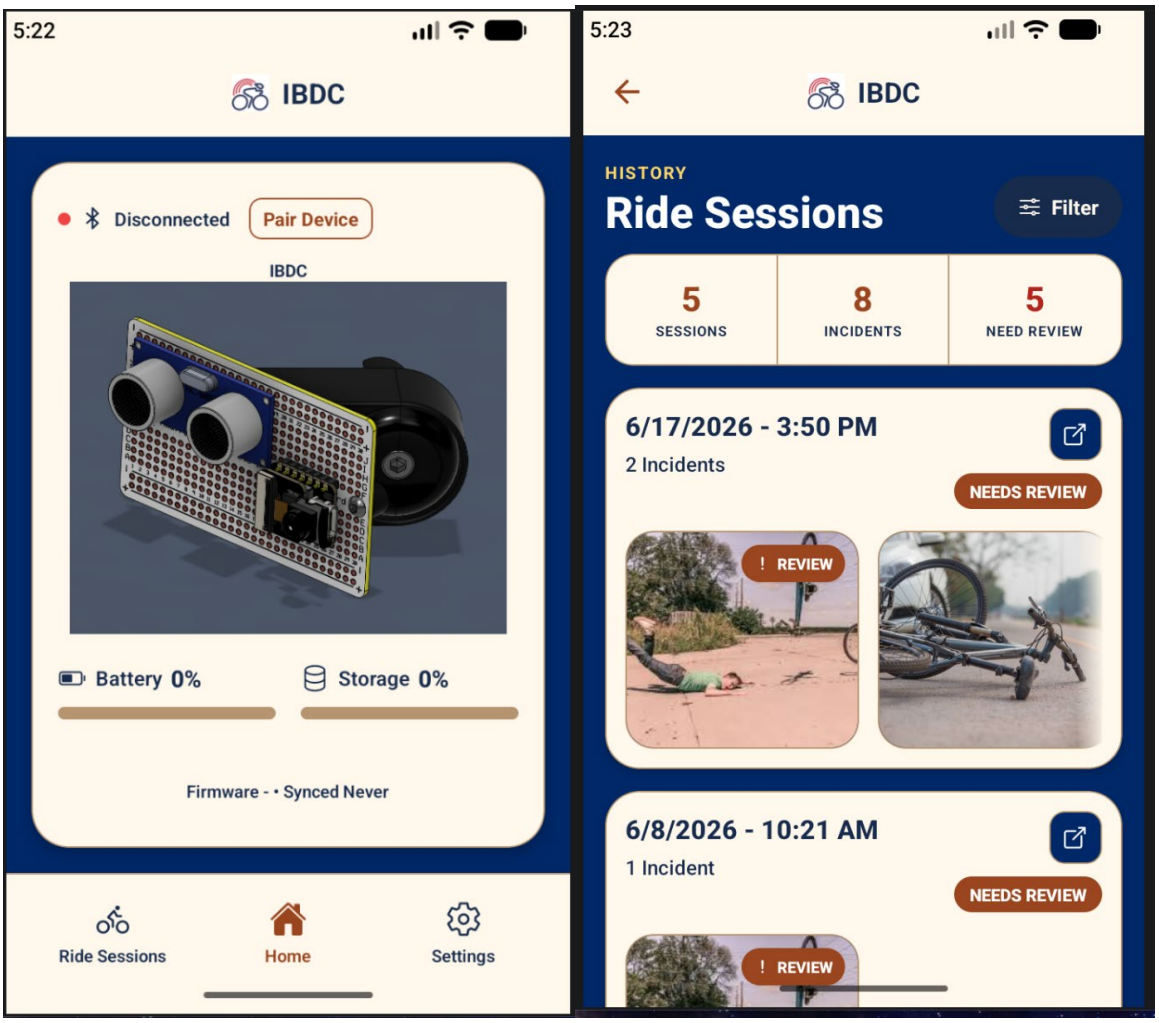
The image we have in the center of the screen on the main page I believe needs a discussion. It wont scale well to fill space and it was always intended as a placeholder I believe. What should we have there knowing that the space it needs to fill may vary?

My suggestion is a state-aware status screen. When there isn't a device connected, we could show a helpful, pair your device to get started message and maybe a link to an online get-started page. While a session is in progress we could show session duration, incident count, images taken count. At all times we could show a counter for incidents needing review. I have not built this yet pending thoughts of the team and sponsor.

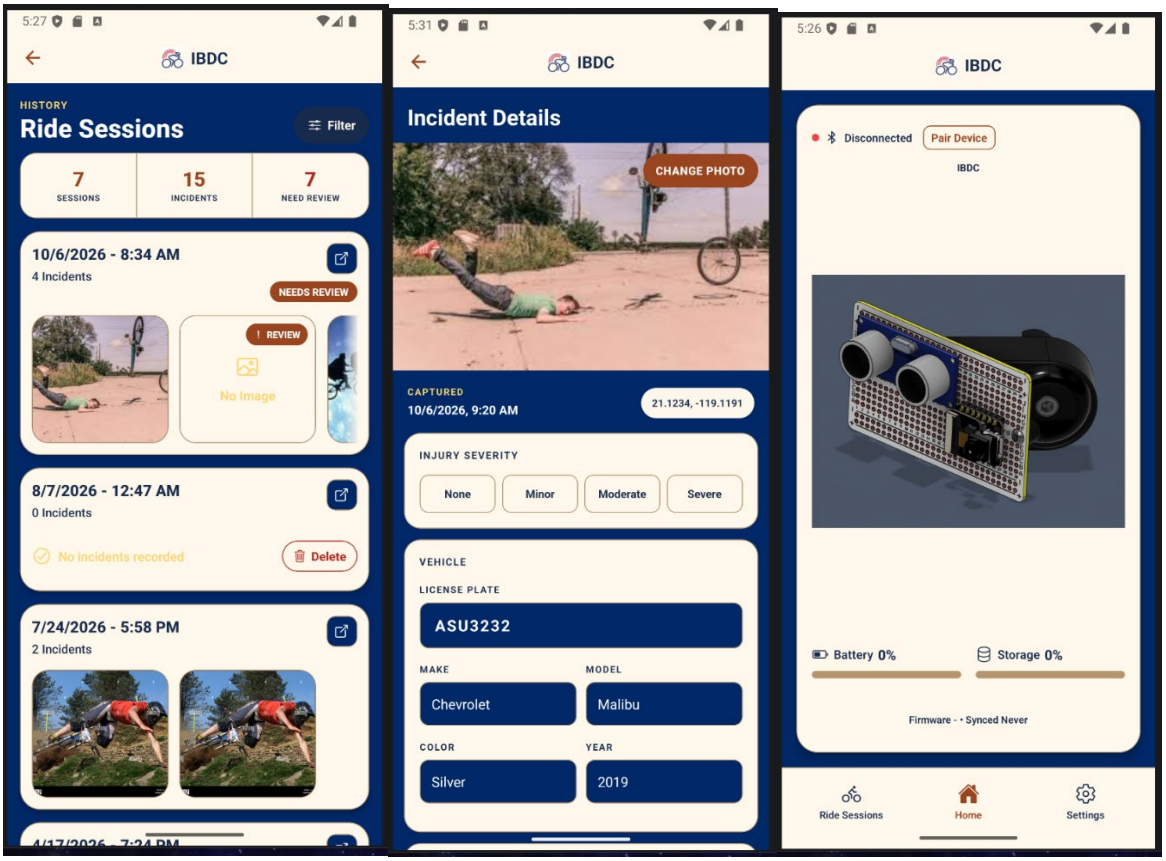
Sessions per page default in the settings screen was 10 but the in-code default was 10. I fixed that.

**On the following 3 pages are images of the updated UI.**

Small Phone Screens (720 x 1280)



Medium Phone Screens (1080 x 2400)



Large Phone Screens (1440 x 3088)

